

1)

```
c = float(input("Enter temperature in Celsius :"))
```

```
# calculate fahrenheit
```

```
fahrenheit = (c* 1.8) + 32
```

```
print('%0.1f degree Celsius is equal to %0.1f degree Fahrenheit'  
%(c,fahrenheit))
```

```
f = float(input("Enter temperature in Fahrenheit :"))
```

```
# calculate Celsius
```

```
celsius = (f - 32) / 1.8
```

```
print('%0.1f degree Fahrenheit is equal to %0.1f degree Celsius'  
%(f,celsius))
```

2a)

```
N=int(input('Enter the number of elements:'))
```

```
def fib(n):
```

```
    if n<=1:
```

```
        return n
```

```
    else:
```

```
        return fib(n-1)+fib(n-2)
```

```
if N<=0:
```

```
    print("Enter the positive number !")
```

```
else:
```

```
    print("The fibbonaci sereies is :")
```

```
    for i in range(N):
```

```
        print(fib(i))
```

2b)

```
N=int(input('Enter the number :'))
```

```
def fact(n):
```

```
    if n==0 or n==1:
```

```
        return 1
```

```
    else:
```

```
        return n*fact(n-1)
```

```
res=fact(N)
```

```
print("The result is :",res)
```

3)

```
s=input("Enter a number:")
```

```
d={}
```

```
print("frequency:")
```

```
for ch in s:
```

```
    d[ch]=d.get(ch,0)+1
```

```
print(d)
```

4)

```
import numpy as np
```

```
lst = []
```

```
# number of elements as input
```

```
n = int(input("Enters number of elements : "))
```

```
print("Enter the elements of the list:")
```

```
# iterating till the range
```

```
for i in range(0, n):
```

```
    ele = int(input())
```

```
    lst.append(ele) # adding the element
```

```
print(lst)
```

```
# Calculating average using average()
```

```
print("Mean=", np.average(lst))
```

```
# Calculating variance using var()
```

```
print("Variance=", np.var(lst))
```

```
# Calculating standard deviation
```

```
print("Standard Deviation=", np.std(lst))
```

5)

empty dictionary

my_dict = {}

number of elements

n = int(input("Enter number of elements: "))

taking input from user

for i in range(n):

key = input("Enter key: ")

value = input("Enter value: ")

my_dict[key] = value

sorting dictionary based on keys

sorted_dict = dict(sorted(my_dict.items()))

displaying results

print("Original Dictionary:", my_dict)

print("Dictionary sorted by keys:", sorted_dict)

6)

```
n = int(input('Enter the number of elements: '))
```

```
a = []
```

```
for i in range(n):
```

```
    m = int(input('Enter the element: ')) # convert to int
```

```
    a.append(m)
```

```
print('The list is:', a)
```

```
file = open("even.txt", "w") # open file in write mode
```

```
for num in a:
```

```
    if num % 2 == 0:
```

```
        file.write(str(num) + "\n")
```

```
file.close()
```

```
print('The file saved successfully!')
```